

COMPASS_Synop- tic_LE_Tree_CO2_Flux_Calcs_2023b

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1 Add Required Packages

2 Pull in Raw Datafiles

```
#defining where the raw data is

DATA_DIR_ROOT <- here::here(
  "Tree_Fluxes_Synoptic_LE",
  "2023",
  "Raw Data"
)

#function to read data
now <- function() format(Sys.time(), "%Y-%m-%d %H:%M:%S")
message(now(), " | Script started")
#finding the zip files
message(now(), " | Looking for ZIP files in: ", DATA_DIR_ROOT)

zip_files <- list.files(
  DATA_DIR_ROOT,
  pattern = "\\\\.zip$",
  full.names = TRUE)

message(now(), " | ZIP files found: ", length(zip_files))
```

```

print(zip_files)
  if (length(zip_files) == 0) {
    stop("No ZIP files found. Check DATA_DIR_ROOT.")}

cache <- list()

# Reader function
read_li7810_zip <- function(zip_file, tz = "US/Eastern") {
  message(now(), " | Processing ZIP: ", basename(zip_file))
  if (zip_file %in% names(cache)) {
    message(" -> Using cached version")
    return(cache[[zip_file]]) }

  tmp_dir <- tempfile("li7810_")
  dir.create(tmp_dir)
  on.exit(unlink(tmp_dir, recursive = TRUE), add = TRUE)
  unzip(zip_file, exdir = tmp_dir)
  data_files <- list.files(
    tmp_dir,
    pattern = "\\\\.data$",
    full.names = TRUE,
    recursive = TRUE )

  message(" -> .data files found: ", length(data_files))
  if (length(data_files) == 0) {
    stop("No .data files inside ZIP: ", zip_file) }

  dat_list <- lapply(data_files, function(f) {
    message("      Reading: ", basename(f))
    ffi_read_LI7810(f)})

  dat <- data.table::rbindlist(dat_list, fill = TRUE) %>%
    mutate(TIMESTAMP = with_tz(TIMESTAMP, tz))
  cache[[zip_file]] <- dat
  message(" -> ZIP finished: ", basename(zip_file))
  return(dat)}

# ACTUAL EXECUTION
message(now(), " | Starting data read")
alldat_list <- lapply(zip_files, read_li7810_zip)
alldat <- data.table::rbindlist(alldat_list, fill = TRUE)
message(now(), " | DONE")

```

```

message("Final number of rows: ", nrow(alldat))

# Show result
print(head(alldat))

#REMOVING WEIRD VERY LOW VALUES ON "2022-12-20 11:41:00" co2=-4000 licor bug???
alldat<-alldat%>%filter(CO2>-1000)

```

3 Read in metadata file and match to raw data

```

md <- "Raw Data/COMPASS_SynopticLE_TreeFlux_Metadata_2023.csv"
dat <- read.csv(md)

metadat <- dat %>%
  # Remove rows without date
  filter(!is.na(Date)) %>%
  # Keep only desired instruments
  filter(SN %in% c("TG10-01448", "TG10-01426", "TG10-01028")) %>%
  # Normalize Plot labels
  mutate(
    Plot_raw = toupper(Plot),
    # Extract plot components
    #Site = str_extract(Plot_raw, "[A-Z]{3}"),
    #Zone = str_extract(Plot_raw, "(TR|UP)"), # somehow all of the UP PTR measurements turned
    Site = toupper(Site),
    Zone = toupper(Zone),
    Replicate = str_extract(Plot_raw, "(?<=TR|UP)[0-9]+"),
    # Letter ONLY if it comes AFTER a number
    Letter = str_extract(Plot_raw, "(?<=[0-9])[A-Z]"),
    # Time normalization
    Time_start = str_replace_all(as.character(Time_start), "\\.", ":") |> str_trim(),
    Stop.time = str_replace_all(as.character(Stop.time), "\\.", ":") |> str_trim(),
    # Safe time parsing (NO WARNINGS)
    t_start = suppressWarnings(parse_date_time(Time_start, orders = c("H:M:S", "H:M"))),
    t_stop = suppressWarnings(parse_date_time(Stop.time, orders = c("H:M:S", "H:M"))),
    # Observation length
    Obs_length = as.numeric(t_stop - t_start, units = "secs"),
    # Timestamp
    TIMESTAMP = as.POSIXct(

```

```

    paste(Date, Time_start),
    format = "%m/%d/%Y %H:%M:%S",
    tz = "EST"
  ),
  # Sequential ID
  Seq = row_number(),
  # INAL standardized Plot label
  Plot = if_else(
    is.na(Letter),
    paste(Date, Site, Zone, Replicate, Seq, sep = "_"),
    paste(Date, Site, Zone, Replicate, Letter, Seq, sep = "_")) )

# Quick check
head(metadat$Plot)

```

4 Check start times overlaps or missing values

```

#checking if there is any start and stop time overlap from the field notes
overlaps_detected <- metadat %>%
  mutate(
    start_ts = as.POSIXct(
      paste(Date, Time_start),
      format = "%m/%d/%Y %H:%M:%S",
      tz = "EST"
    ),
    end_ts = start_ts + Obs_length
  ) %>%
  arrange(SN, start_ts) %>%
  group_by(SN) %>%
  mutate(
    next_plot      = lead(Plot),
    next_start_ts  = lead(start_ts),
    overlap_fffi   = !is.na(next_start_ts) & next_start_ts <= end_ts
  ) %>%
  filter(overlap_fffi) %>%
  select(
    SN,
    Plot,
    start_ts,

```

```

    end_ts,
    next_plot,
    next_start_ts,
    Obs_length
  )

print(overlaps_detected)

#checking problems with obs_length - this is a frequent bug for manual field notes
field_notes_overlap<-metadat %>%
  mutate(
    obs_problem =
      is.na(Obs_length) | Obs_length <= 0
  ) %>%
  filter(obs_problem) %>%
  select(
    SN,
    Date,
    Plot,
    Time_start,
    Stop.time,
    Obs_length
  )

print(field_notes_overlap)

# checking unique measurment start time and stop time
# alldat %>%
#   filter(
#     SN == "TG10-01448",
#     TIMESTAMP > as.POSIXct("2023-01-05 13:28:00"),
#     TIMESTAMP < as.POSIXct("2023-01-05 13:33:00")
#   ) %>%
#   ggplot(aes(x = TIMESTAMP, y = CO2)) +
#   geom_point()#ylim(300,500)

```

5 Match metadata to raw data

```

#function to match data per equipment (considering different serial numbers) uhul!
match_flux_metadata <- function(alldat, metadat) {

```

```

message("Starting metadata  Licor matching")
#Sanity checks (fail early)
stopifnot(
  "TIMESTAMP" %in% names(alldat),
  "SN"         %in% names(alldat),
  "Date"       %in% names(metadat),
  "Time_start"%in% names(metadat),
  "Obs_length"%in% names(metadat),
  "Plot"       %in% names(metadat)
)

stopifnot(all(metadat$Obs_length > 0, na.rm = TRUE))
stopifnot(anyDuplicated(metadat$Plot) == 0)
# Instruments present in BOTH datasets
instruments <- intersect(
  unique(alldat$SN),
  unique(metadat$SN)
)
message("Instruments to process: ", paste(instruments, collapse = ", "))
results <- lapply(instruments, function(sn) {
  message("Processing SN: ", sn)
  dat_i <- alldat %>%
    filter(SN == sn)
  meta_i <- metadat %>%
    filter(SN == sn) %>%
    arrange(Date, Time_start) %>%
    mutate(metadat_row = row_number())
  stopifnot(nrow(meta_i) > 0)
# Actual matching
dat_i$metadat_row <- ffi_metadata_match(
  data_timestamps = dat_i$TIMESTAMP,
  start_dates     = meta_i$Date,
  start_times     = meta_i$Time_start,
  obs_lengths     = meta_i$Obs_length )
# Attach metadata (vectorized, fast)
dat_i <- dat_i %>%
  mutate(
    Plot      = meta_i$Plot[metadat_row],
    Site      = meta_i$Site[metadat_row],
    Zone      = meta_i$Zone[metadat_row],
    Volume    = meta_i$Volume[metadat_row],
    Temp      = meta_i$Temp[metadat_row],

```

```

      Area      = meta_i$Area[metadat_row],
      Dead_band = meta_i$Dead_band[metadat_row]      )
# Drop unmatched rows
n_before <- nrow(dat_i)
dat_i <- dat_i %>% filter(!is.na(metadat_row))
n_after <- nrow(dat_i)
message("  Matched rows: ", n_after, " / ", n_before)
dat_i })
alldat_matched <- rbindlist(results, fill = TRUE)
#Final sanity checks
stopifnot(
  !anyNA(alldat_matched$Plot),
  nrow(alldat_matched) > 0
)
message("Matching finished successfully")
return(alldat_matched)}

#running the funciton
message("Running final metadata matching...")
alldat_matched <- match_flux_metadata(alldat, metadat)
message("Matched dataset rows: ", nrow(alldat_matched))
head(alldat_matched)

#saving matched data and metadata as raw data is too big (today, Licor files leave in the CO2 column)
saveRDS(alldat_matched, file = "alldat_matched_2023b.rds")

#can remove any row that have NA's in the CO2 column
#alldat <- alldat %>%
# filter(!is.na(CO2))

```

6 reating alldata matched df

```

# after you match the data for the first time - we can just read the .rds file to save processing
# if you need to match licor data and metadata again, remove "#" from the previous chunks

alldat_matched <- readRDS("alldat_matched_2023b.rds")

```

```

##Extracting initial CO2 concentration

```

```

## Extract initial CO2 concentration (t0) for each Plot for QAQC later
#saving df to use after flux calculation
df_CO2_in <- alldat_matched %>%
  arrange(Plot, TIMESTAMP) %>%
  group_by(Plot) %>%
  summarise(CO2_initial = first(CO2), .groups = "drop")

```

7 Creating temperature csv from the L2-weather station in the wetland

```

# Getting Air temp from the weather station
## Read L2 temperature data from ZIP stored outside GitHub,
## extract desired variable, save lightweight CSV,
## and remove temporary unzipped files afterwards

## Read extracted L2 Parquet files and save temperature time series
BASE_DIR <- "D:/COMPASS_L2-v2-1/v2-1"
SITE <- "(OWC|CRC|PTR)"
VARIABLE <- "wx-tempavg15"

OUTFILE <- "Raw Data/L2_temp_timeseries.csv"

# Build regex that matches REAL L2 filenames
regex <- paste0(SITE, ".*", VARIABLE, ".*_L2_v2-1.*\\.parquet$")

# Locate Parquet files
files <- list.files(BASE_DIR, pattern = regex, recursive = TRUE, full.names = TRUE)

message("Files found: ", length(files))
print(basename(files))

# Fail fast if nothing is found
stopifnot(length(files) > 0)

# Read and combine Parquet files
dat <- lapply(files, function(f) {
  message("Reading ", basename(f))
  read_parquet(f)
})

```

```

dat <- bind_rows(dat)

# Optional QA filtering (only if column exists)
if ("QA_flag" %in% names(dat)) {
  dat <- dat %>% filter(QA_flag == 0)
}

# Keep only necessary columns
temps <- dat %>%
  select(TIMESTAMP, Plot, Site, Value) %>%
  rename(Temp = Value)

head(temps)

# Save lightweight output only
write.csv(
  temps,
  file = OUTFILE,
  row.names = FALSE
)

```

8 Reading Temp df to merge to metadat

```

# reading temp
temps <- read.csv("Raw Data/L2_temp_timeseries.csv")

# make sure timestamps are in the correct format
# convert to tibble early to avoid data.table list-column issues
alldat_matched <- as_tibble(alldat_matched) %>%
  mutate(across(where(is.list), ~ as.numeric(unlist(.))))

alldat_matched$TIMESTAMP <- as.POSIXct(alldat_matched$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")
alldat_matched$Dead_band <- as.numeric(alldat_matched$Dead_band)

# fix midnight timestamps in L2 data
temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)] <-
  paste0(temps$TIMESTAMP[!grepl(":", temps$TIMESTAMP)], " 00:00:00")
temps$TIMESTAMP <- as.POSIXct(temps$TIMESTAMP, format = "%Y-%m-%d %H:%M:%OS")

```

```

# time rounding
alldat_matched$TIMESTAMP_round <- round_date(alldat_matched$TIMESTAMP, "60 mins")
temps$TIMESTAMP_round <- round_date(temps$TIMESTAMP, "60 mins")

# one observation per hour
temps_clean <- temps %>%
  group_by(TIMESTAMP_round, Site) %>%
  summarize(Temp = mean(Temp), .groups = "drop")

# join weather station temp
alldat_matched <- alldat_matched %>%
  left_join(temps_clean, by = c("Site", "TIMESTAMP_round"))

# DIAGNOSTIC: which plots are missing Temp after join
temp_missing_check <- alldat_matched %>%
  group_by(Plot, Site) %>%
  summarise(
    n_obs = n(),
    n_temp_na = sum(is.na(Temp)),
    pct_temp_na = round(100 * n_temp_na / n_obs, 1),
    month = month(first(TIMESTAMP)),
    .groups = "drop"
  ) %>%
  filter(n_temp_na > 0)

message("Plots with missing Temp: ", nrow(temp_missing_check))

```

Plots with missing Temp: 2

```
print(temp_missing_check)
```

```
# A tibble: 2 x 6
```

	Plot <chr>	Site <chr>	n_obs <int>	n_temp_na <int>	pct_temp_na <dbl>	month <int>
1	2/21/2023_CRC_UP_304_68	CRC	720	720	100	2
2	2/21/2023_CRC_UP_305_69	CRC	360	360	100	2

```

# IMPUTATION: monthly mean per site as fallback
monthly_temp <- alldat_matched %>%
  mutate(month = month(TIMESTAMP)) %>%

```

```

filter(!is.na(Temp)) %>%
group_by(Site, month) %>%
summarise(Temp_monthly_mean = mean(Temp), .groups = "drop")

alldat_matched <- alldat_matched %>%
  mutate(
    month      = month(TIMESTAMP),
    Temp       = as.numeric(Temp)
  ) %>%
left_join(monthly_temp, by = c("Site", "month")) %>%
mutate(
  Temp_source = case_when(
    !is.na(Temp)           ~ "weather_station",
    !is.na(Temp_monthly_mean) ~ "monthly_mean_imputed",
    TRUE                  ~ "still_missing"
  ),
  Temp = coalesce(Temp, Temp_monthly_mean)
) %>%
select(-Temp_monthly_mean, -month)

# FINAL CHECK
message("Temp source summary:")

```

Temp source summary:

```

alldat_matched$Temp_source <- as.character(alldat_matched$Temp_source)
table(alldat_matched$Temp_source)

```

monthly_mean_imputed	weather_station
1080	131125

```

still_missing <- alldat_matched %>%
  filter(Temp_source == "still_missing") %>%
  distinct(Plot, Site)

if (nrow(still_missing) > 0) {
  message("WARNING: ", nrow(still_missing), " plots still have no Temp after imputation:")
  print(still_missing)
} else {

```

```
message("All plots have Temp after imputation.")
}
```

All plots have Temp after imputation.

9 fixing some Deadbands and stop time

```
# Table of plots with custom deadband adjustments
time_overrides <- tibble::tibble(
  Plot = c("#7/13/2023_CRC_TR_315_115",
            "#7/27/2023_PTR_UP_8_125",
            "#3/28/2023_OWC_TR_127_170",
            "#5/10/2023_OWC_UP_103_73",
            "#7/13/2023_CRC_TR_314_114",
            "#8/3/2023_CRC_TR_314_133",
            "#7/13/2023_CRC_TR_316_116",
            "#6/29/2023_CRC_TR_311_104" ,
            "2/24/2023_PTR_UP_4_100"),
  dead_band_manual = c(#8,9,22,16,8,16,3,34,
                       22),
  stop_offset_secs = NA_real_ )

#apply new incubation period
alldat_trimmed <- alldat_matched %>%
  left_join(time_overrides, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    t_start_original = min(TIMESTAMP),
    t_end_original   = max(TIMESTAMP),

    # deadband manual or metadat
    t_start_adjusted = if_else(
      !is.na(dead_band_manual),
      t_start_original + dead_band_manual,
      t_start_original + as.numeric(coalesce(first(Dead_band), 0))
    ),

    # new offset
    t_end_adjusted = if_else(
```

```

      !is.na(stop_offset_secs),
      t_end_original + stop_offset_secs,
      t_end_original
    )
  ) %>%
  filter(TIMESTAMP >= t_start_adjusted,
         TIMESTAMP <= t_end_adjusted) %>%
  ungroup() %>%
  select(-t_start_original, -t_end_original,
         -t_start_adjusted, -t_end_adjusted,
         -dead_band_manual, -stop_offset_secs)

# alldat_trimmed %>%
#   filter(Plot == "8/17/2022_OWC_UP_6_147") %>%
#   ggplot(aes(TIMESTAMP, CO2)) +
#   geom_point() + theme_classic()

```

10 Calculate fluxes

```

#change the units
alldat_trimmed$CO2_umol <- ffi_ppm_to_umol(alldat_trimmed$CO2,
                                           volume = alldat_trimmed$Volume, # m3
                                           temp = alldat_trimmed$Temp)      # degrees C from the weather

```

Assuming atm = 101325 Pa

Using R = 8.31446261815324 m3 Pa K-1 mol-1

```

#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1

alldat_trimmed$CO2_umol_m2 <- alldat_trimmed$CO2_umol / 0.16

fluxes <- ffi_compute_fluxes(alldat_trimmed ,
                             group_column = "Plot",
                             time_column = "TIMESTAMP",
                             gas_column = "CO2_umol_m2",
                             dead_band = 0)

```


[illegible]

[illegible]

	HM81_p.value	HM81_r.squared	HM81_RMSE	lin_AIC	lin_flux.estimate
1	7.666169e-156	0.6938552	0.05980751	-3046.562	0.001226571
2	4.580077e-198	0.6957943	0.18877166	-3577.708	0.003108942
3	1.274356e-142	0.6610713	0.11633224	-3306.504	0.002296731
4	7.782742e-168	0.7208468	0.18724805	-1774.056	-0.004036002
5	1.376542e-143	0.6635825	0.06473018	-3148.011	0.001270600
6	4.004211e-161	0.7060499	0.12301382	-2749.865	0.002600406

	lin_flux.std.error	lin_int.estimate	lin_int.std.error	lin_p.value
1	8.976872e-06	15.63384	0.001550952	0
2	7.582276e-06	15.30115	0.001660215	0
3	7.228508e-06	15.50874	0.001248884	0
4	2.592151e-05	19.25761	0.004478512	0
5	8.249156e-06	15.54712	0.001425223	0
6	1.149485e-05	15.44359	0.001985988	0

	lin_r.squared	lin_RMSE	poly_AIC	poly_r.squared	poly_RMSE	rob_AIC
1	0.9689635	0.01904272	-3200.647	0.9761523	0.01672027	-3046.237
2	0.9955116	0.02292967	-4132.399	0.9978481	0.01589799	-3575.361
3	0.9941114	0.01533390	-3317.696	0.9942586	0.01516638	-3306.277
4	0.9759267	0.05498753	-2564.355	0.9935938	0.02841353	-1774.043
5	0.9754138	0.01749901	-3216.570	0.9782143	0.01649987	-3147.846
6	0.9884500	0.02438412	-3222.624	0.9947822	0.01641684	-2749.373

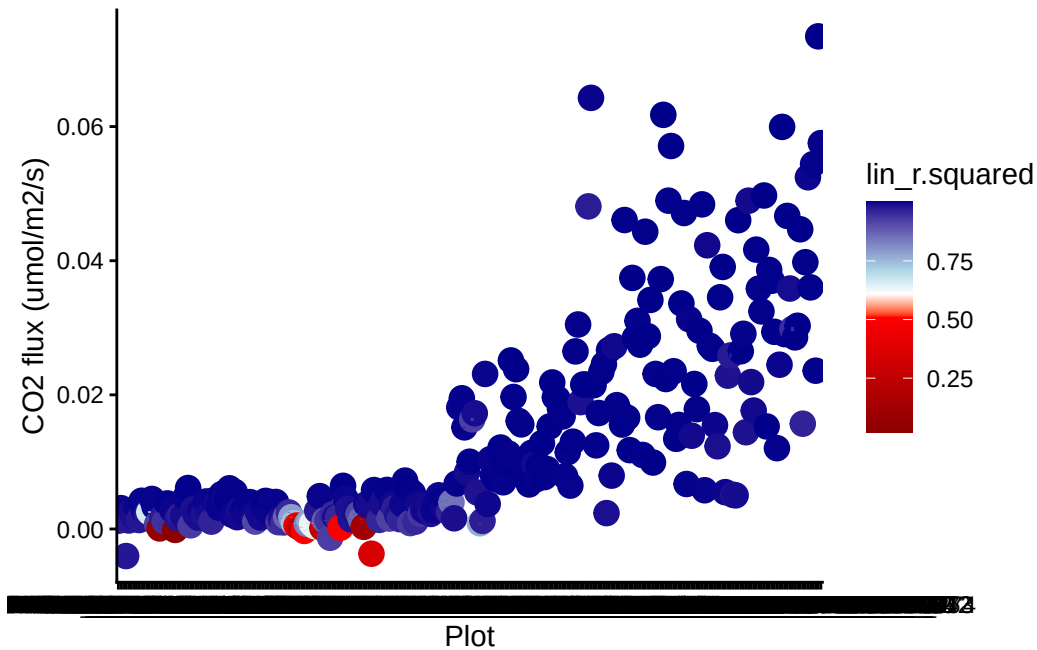
	rob_converged	rob_flux.estimate	rob_flux.std.error	rob_RMSE
1	TRUE	0.001221808	9.771540e-06	0.02051275
2	TRUE	0.003119893	7.855219e-06	0.02148676
3	TRUE	0.002299814	7.665480e-06	0.01546289
4	TRUE	-0.004034655	2.673182e-05	0.06949846
5	TRUE	0.001269843	8.180208e-06	0.01602977
6	TRUE	0.002594641	1.180629e-05	0.02575103

	TIMESTAMP_min		TIMESTAMP_max	
1	2023-01-12 11:23:00	2023-01-12 11:27:59		
2	2023-01-12 11:38:40	2023-01-12 11:44:59		
3	2023-01-12 11:53:00	2023-01-12 11:57:59		
4	2023-01-12 12:05:00	2023-01-12 12:09:59		
5	2023-01-12 12:16:00	2023-01-12 12:20:59		
6	2023-01-12 12:24:00	2023-01-12 12:28:59		

```
#merging fluxes with initial CO2 concentration
fluxes <- fluxes %>%
  left_join(df_CO2_in,by = "Plot")

#take a look at the fluxes we calculated with R2
ggplot(fluxes, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
```

```
geom_point(size=4) + theme_classic() +
geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                  ymax = lin_flux.estimate+ lin_flux.std.error)) +
scale_color_gradientn(colours = c("darkred", "red", "white", "lightblue", "darkblue"),
                      values = c(0, 0.5, 0.6, 0.7, 1)) +
ylab("CO2 flux (umol/m2/s)")
```



```
#sort fluxes by R2
fluxes_good <- as.data.frame(subset(fluxes, lin_r.squared >= .90, select = Plot:TIMESTAMP_ma
fluxes_lowr2 <- as.data.frame(subset(fluxes, lin_r.squared < .90 & lin_flux.estimate < 1, se
fluxes_weird <- as.data.frame(subset(fluxes, lin_r.squared < .90, select = Plot:TIMESTAMP_ma

#find fluxes that represent the first and fourth quartile of the data
low <- quantile(fluxes$lin_flux.estimate, 0.05)
high <- quantile(fluxes$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(fluxes, lin_flux.estimate <= low, select = Plot:TIMESTAMP
fluxes_high <- as.data.frame(subset(fluxes, lin_flux.estimate >= high, select = Plot:TIMESTAI

tails <- rbind(fluxes_low, fluxes_high)
```

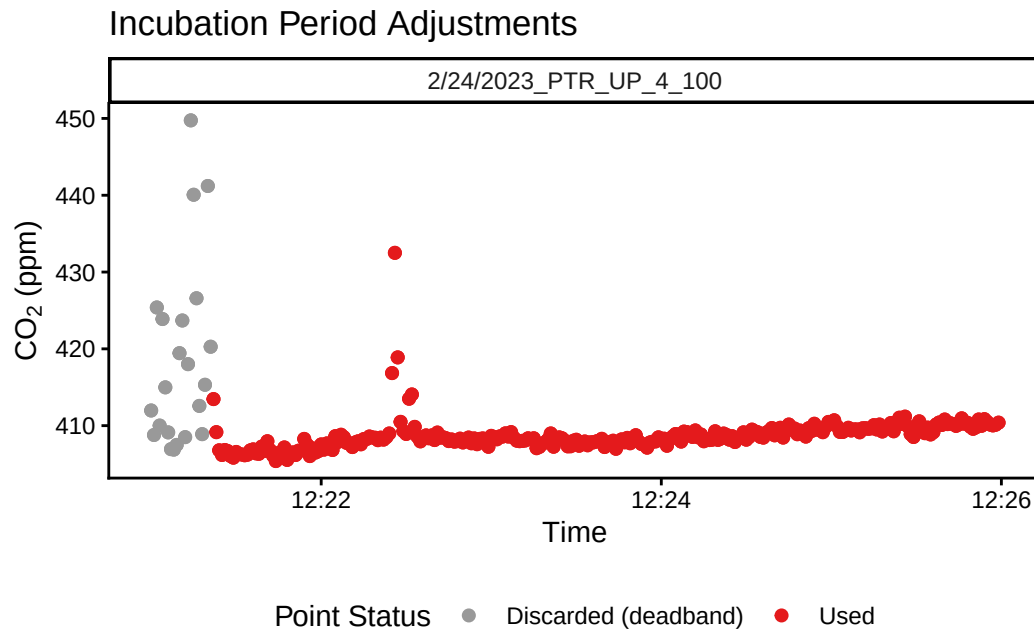
11 check the fluxes for which I changed the start time due to lags or high values (recorded in notes of metadata)

```
# grab trim limits for each overridden Plot
trim_limits <- alldat_trimmed %>%
  filter(Plot %in% time_overrides$Plot) %>%
  group_by(Plot) %>%
  summarise(
    t_start_trim = min(TIMESTAMP),
    t_end_trim   = max(TIMESTAMP) )

# use full alldat_matched + trim limits to classify each point
r_all <- alldat_matched %>%
  filter(Plot %in% time_overrides$Plot) %>%
  left_join(trim_limits, by = "Plot") %>%
  group_by(Plot) %>%
  mutate(
    time_sec = as.numeric(difftime(TIMESTAMP, min(TIMESTAMP), units = "secs")),
    point_status = case_when(
      TIMESTAMP < t_start_trim ~ "Discarded (deadband)",
      TIMESTAMP > t_end_trim   ~ "Discarded (stop trim)",
      TRUE                     ~ "Used" )) %>%
  ungroup()

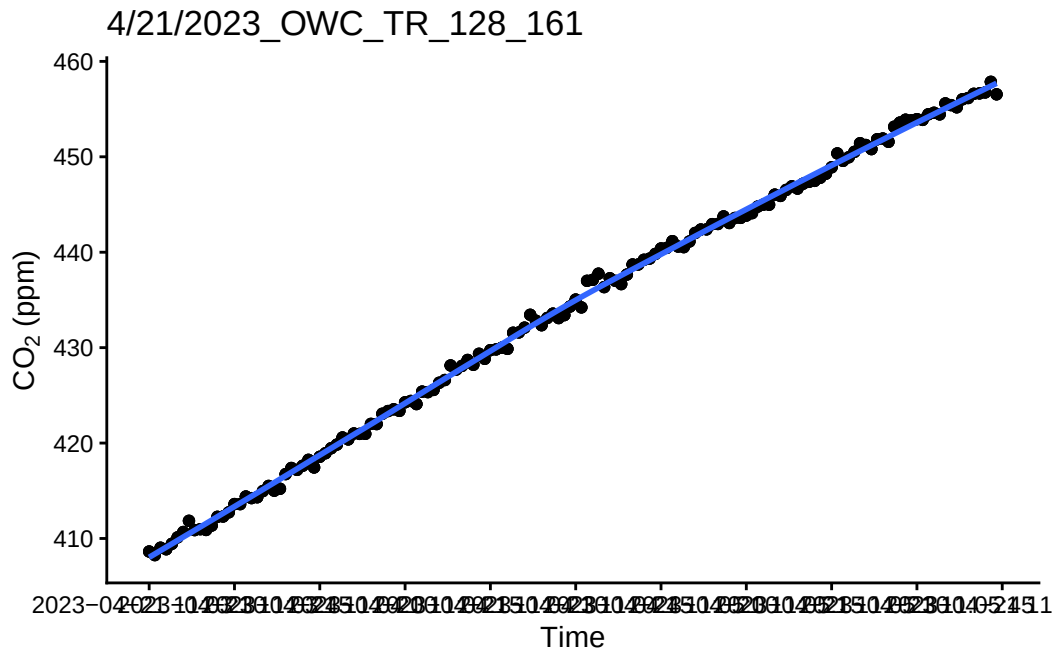
# plot all overridden plots
ggplot(r_all, aes(x = TIMESTAMP, y = CO2, color = point_status)) +
  geom_point(size = 1.8) +
  facet_wrap(~ Plot, scales = "free", ncol = 2) +
  scale_color_manual(
    values = c(
      "Used" = "#E41A1C",
      "Discarded (deadband)" = "grey60",
      "Discarded (stop trim)" = "grey20"
    )
  ) +
  theme_classic() +
  ylab(expression(paste(CO[2], " (ppm)"))) +
  xlab("Time") +
  labs(
    title = "Incubation Period Adjustments",
    color = "Point Status") +
```

```
theme(legend.position = "bottom")
```

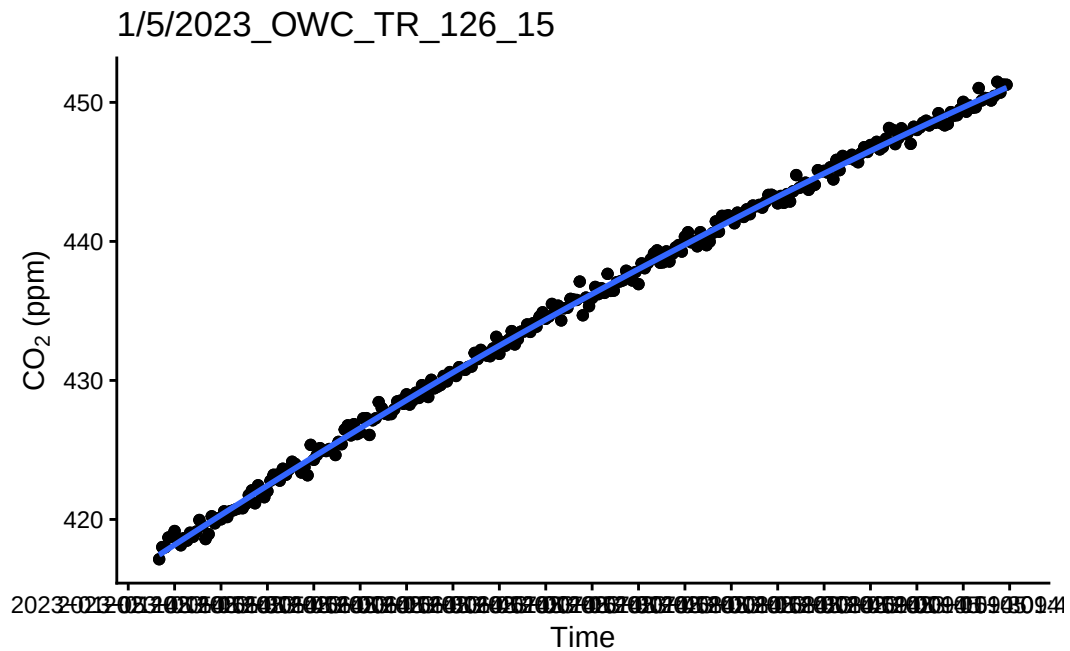


12 Spot check a few fluxes with $R^2 > 0.9$

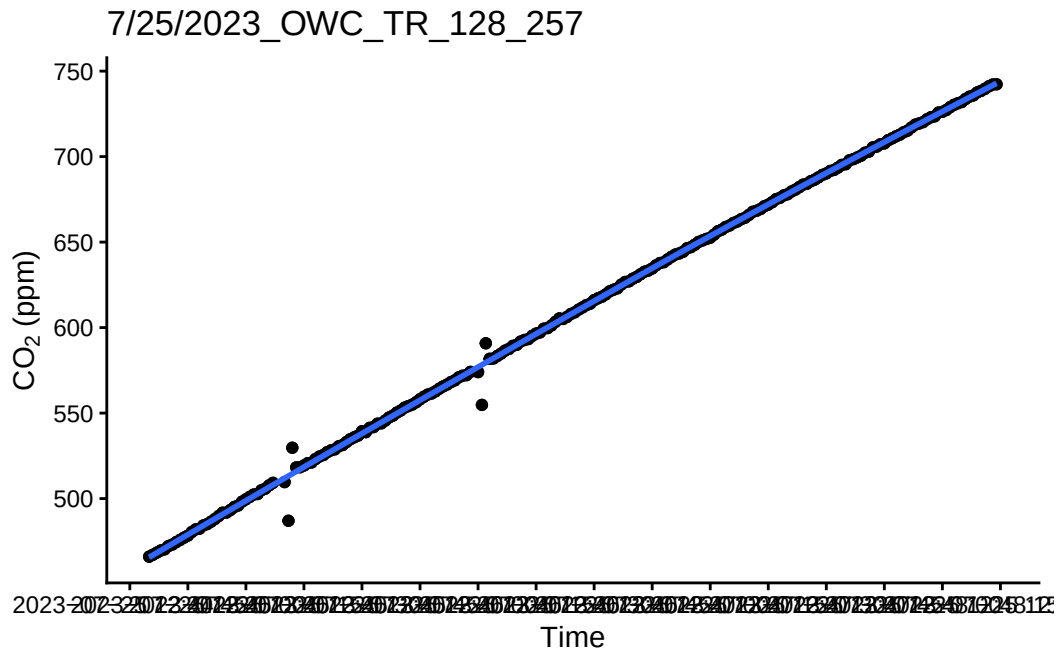
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



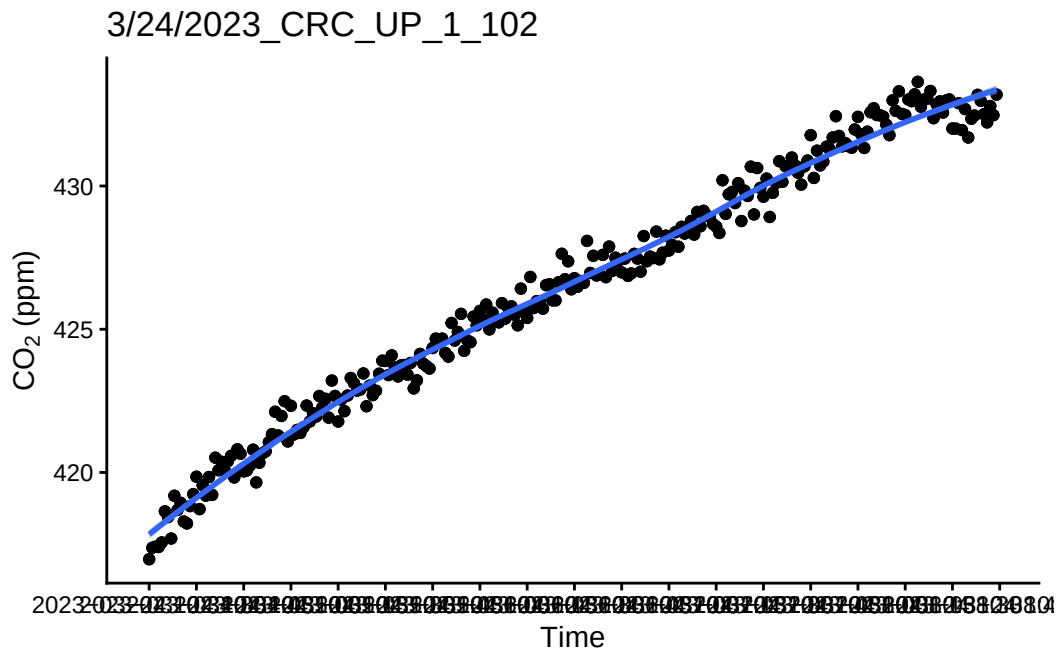
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



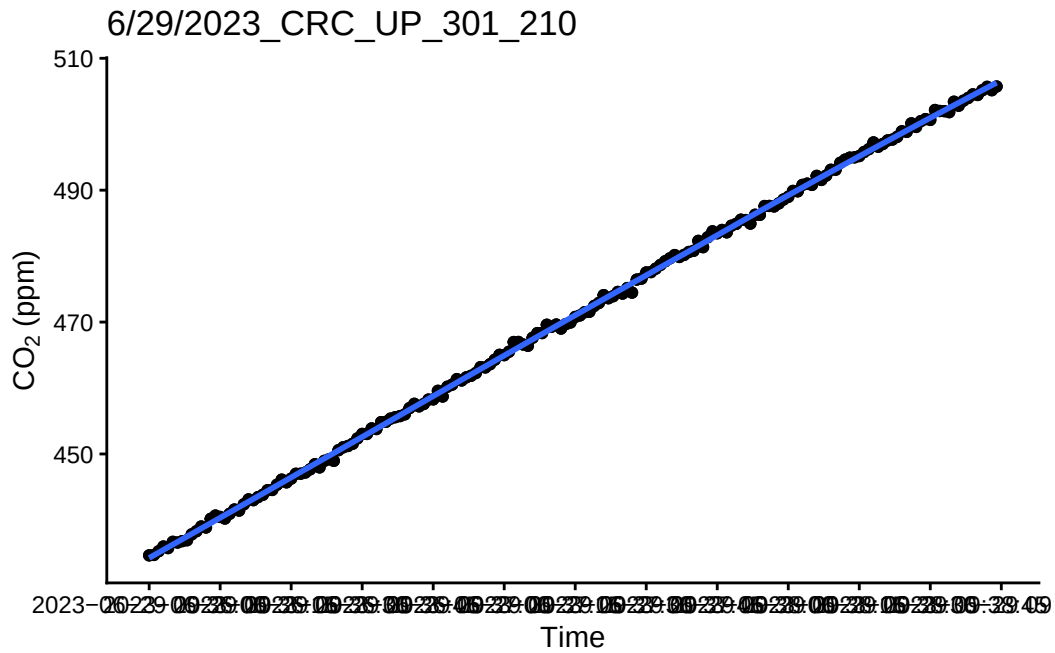
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



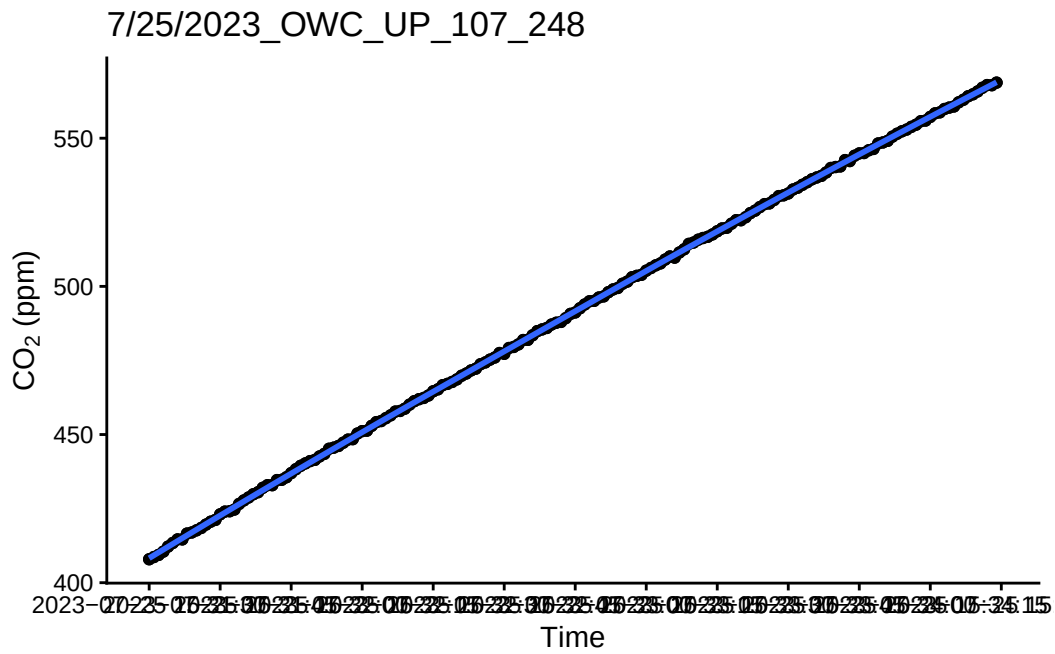
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



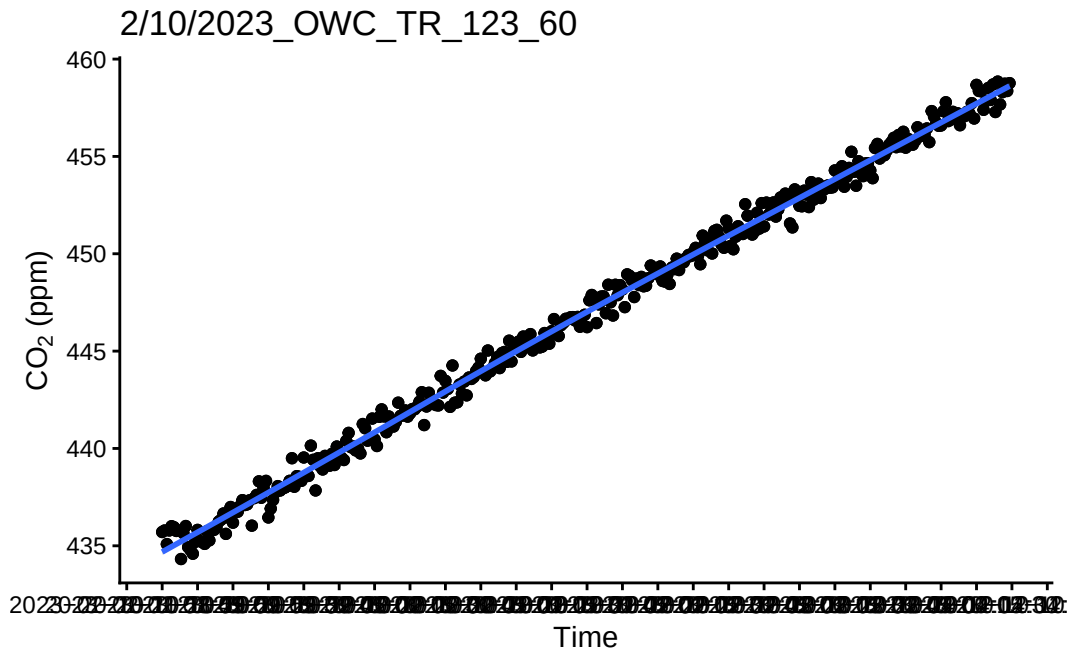
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



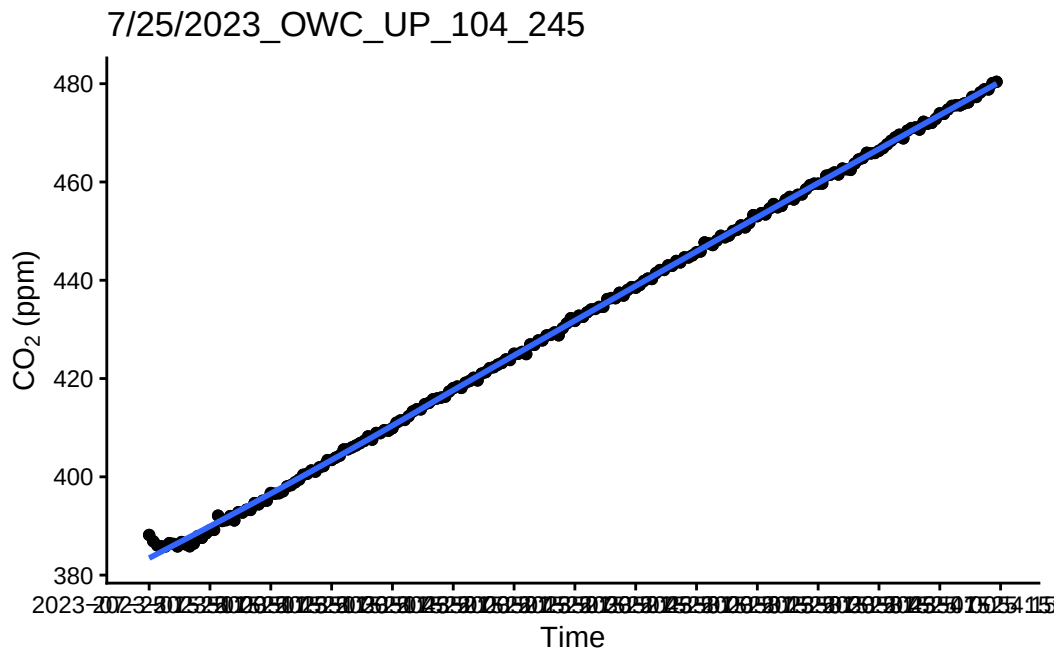
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



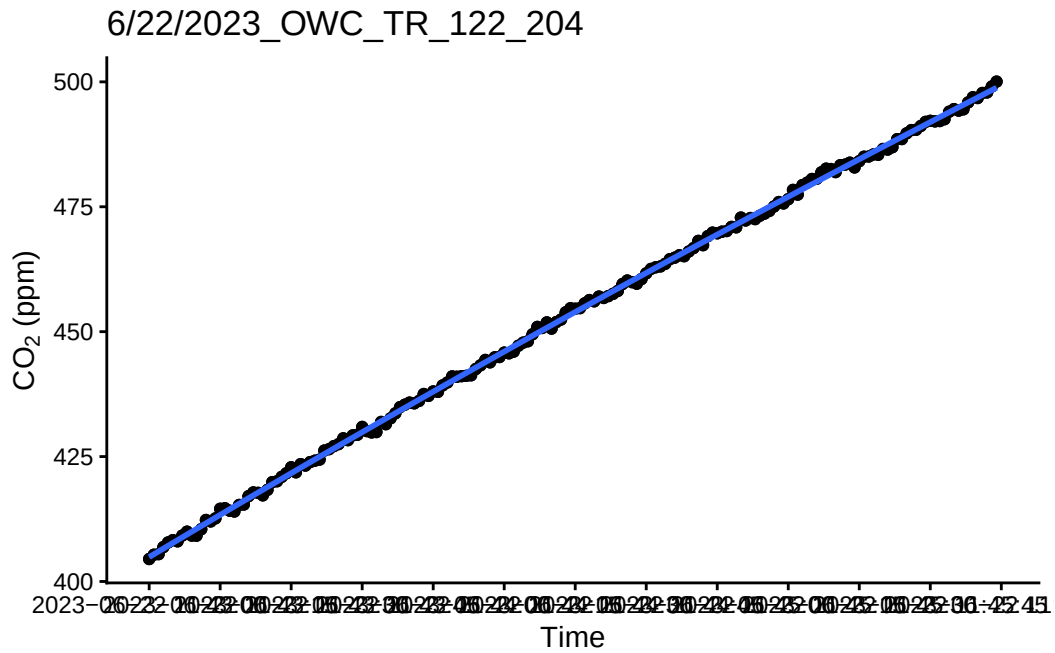
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

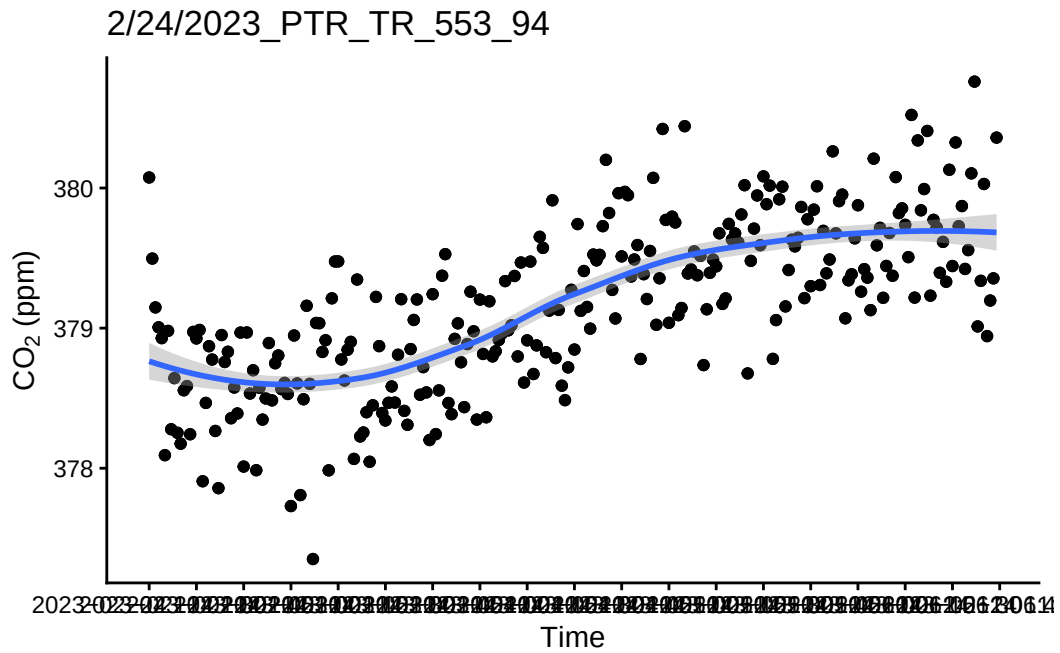


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

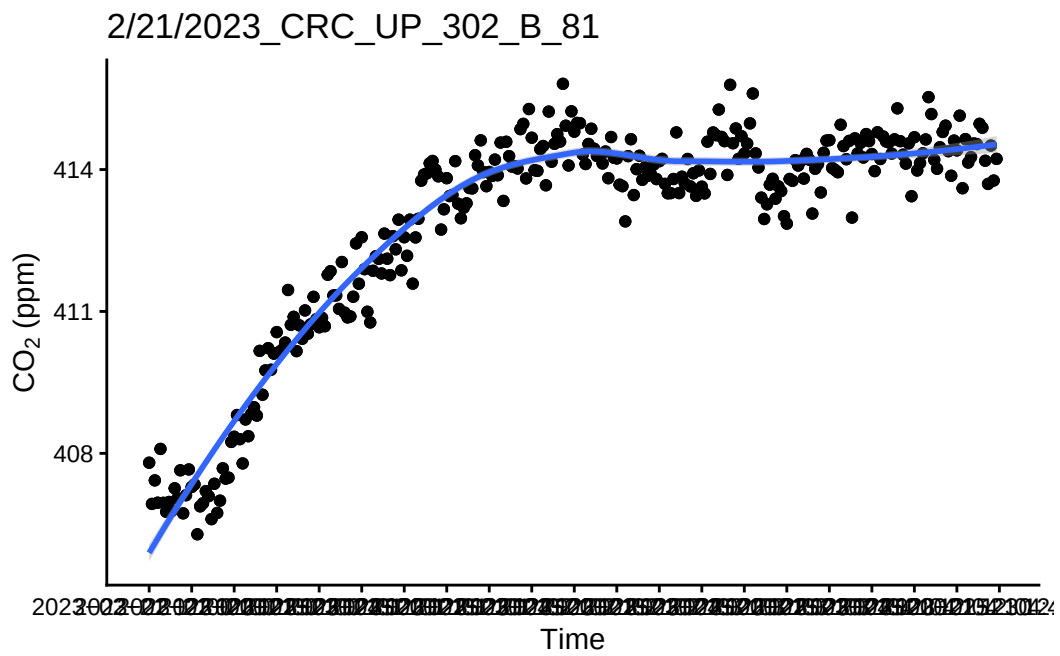


13 Take a look at the fluxes <0.9 and a flux estimate below 1

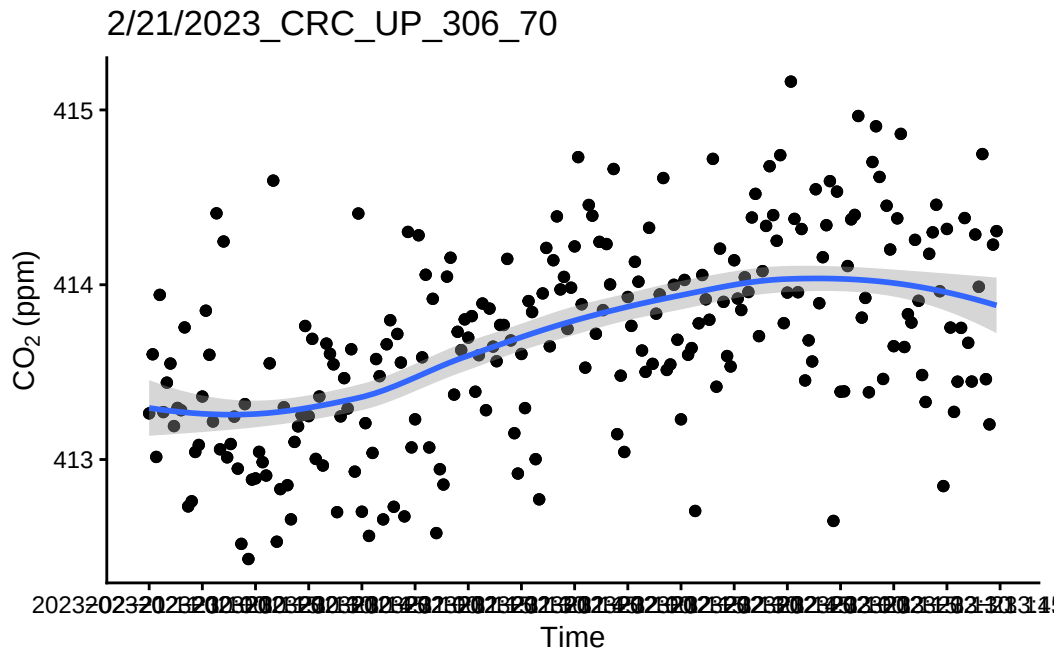
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



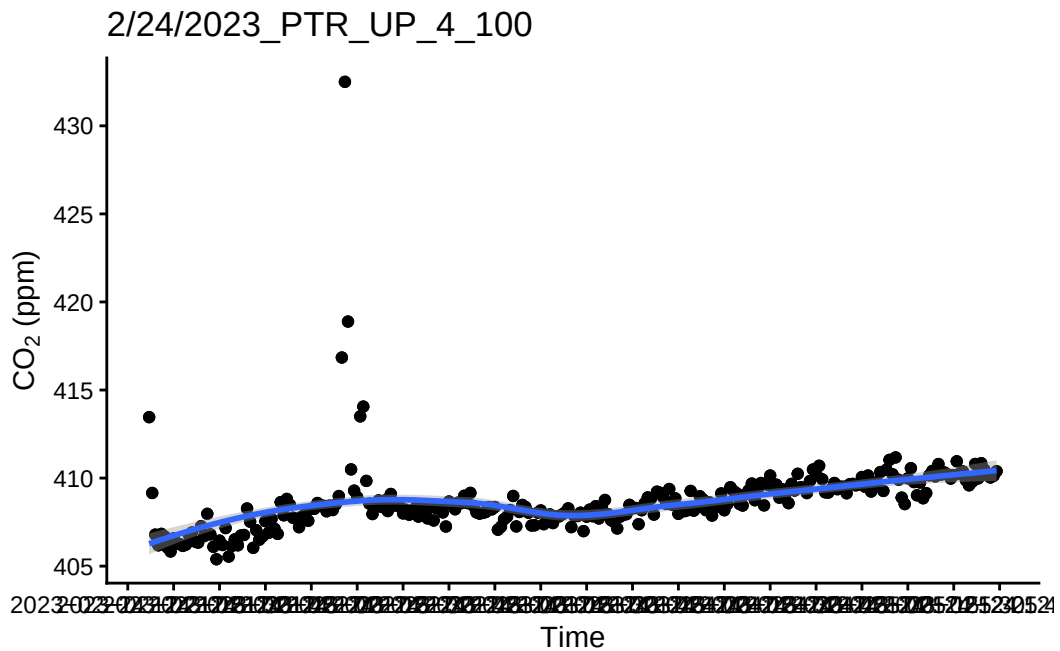
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



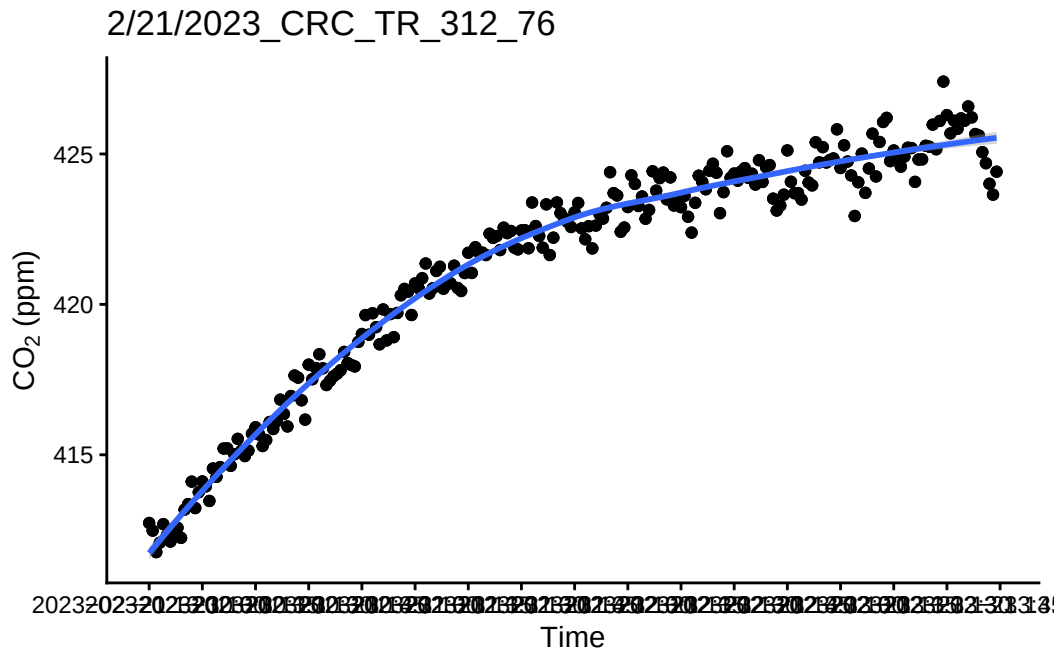
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



14 Take a look at the bad fluxes <0.9 and have a high flux estimate

15 See if we want to remove these fluxes or if we want to try and recalculate these ones?

```
### pull out fluxes with R2 < 0.9 and put them in a list
bad_plots <- unique(fluxes_weird$Plot)

# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
bad_sample <- sample(bad_plots, 5, replace=FALSE)

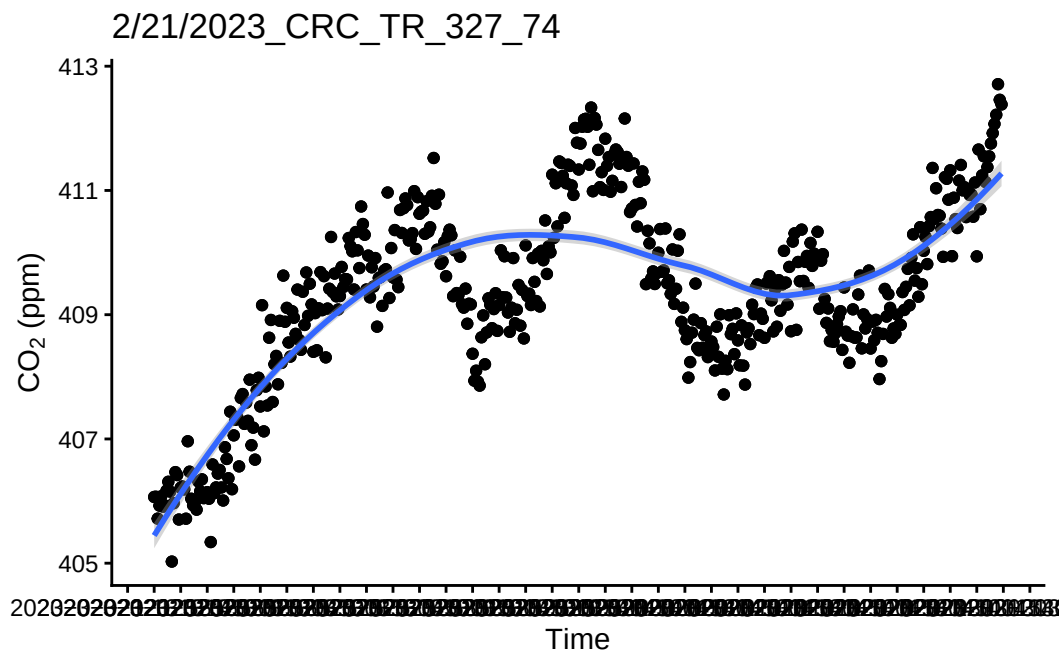
#Loop to run five random plots subset above
for(P in bad_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste(CO[2], " (ppm)"))) +
```

```

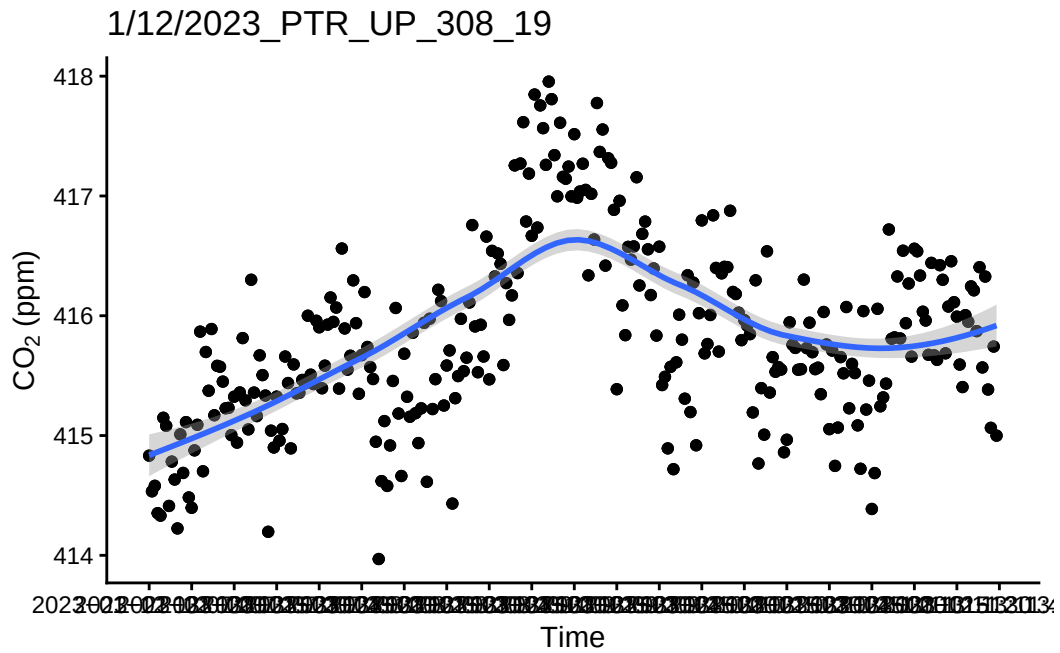
xlab("Time") + labs(title = P) +
geom_smooth()
print(p1)
}

```

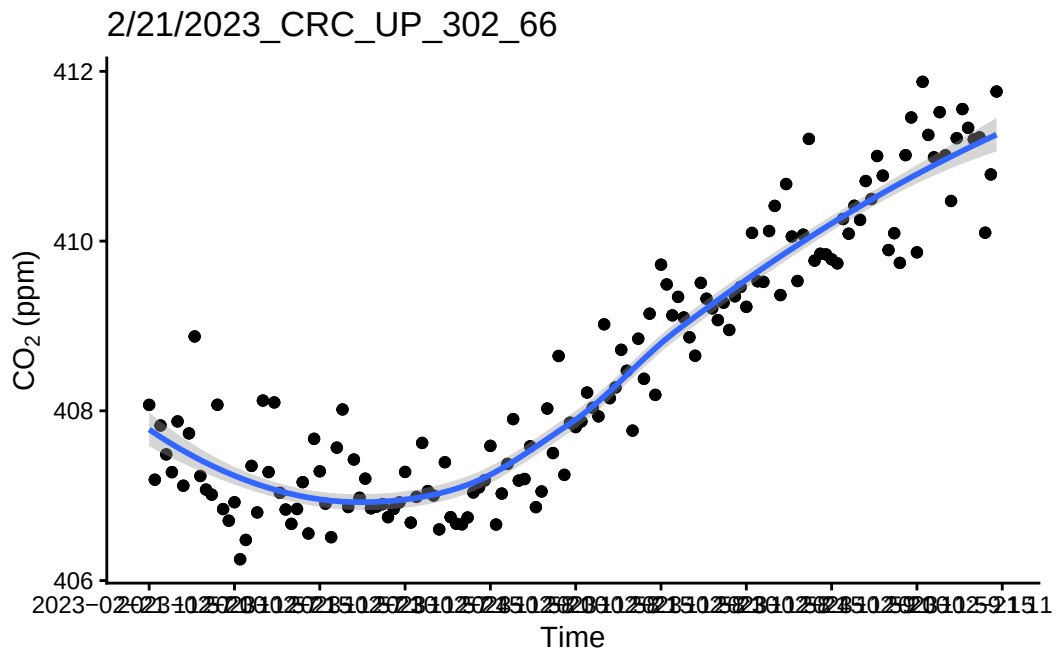
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



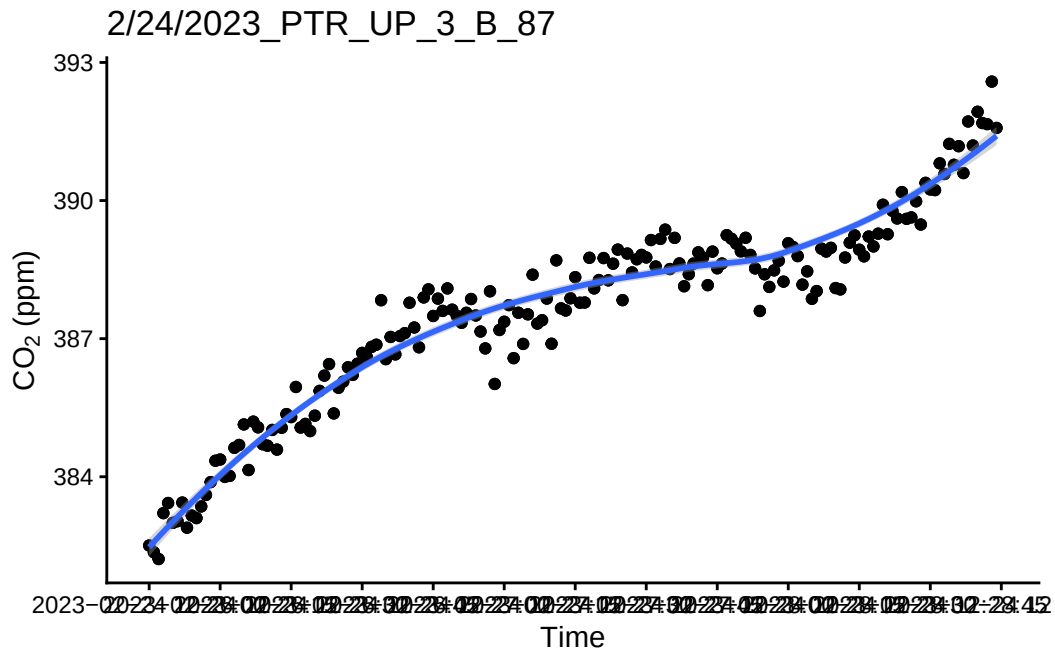
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



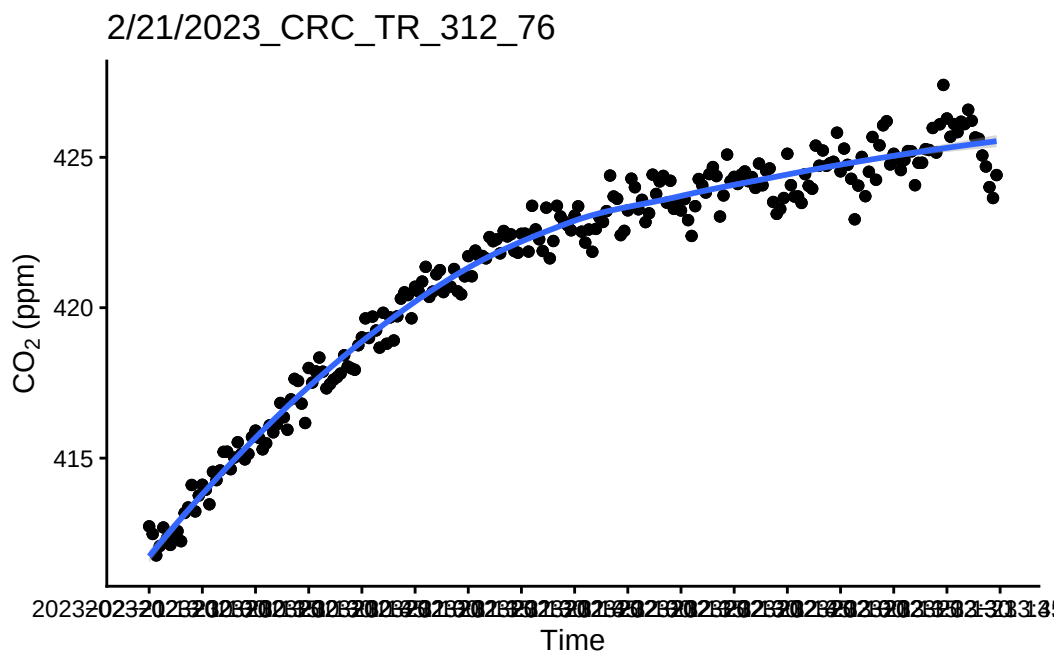
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



```
#####
```

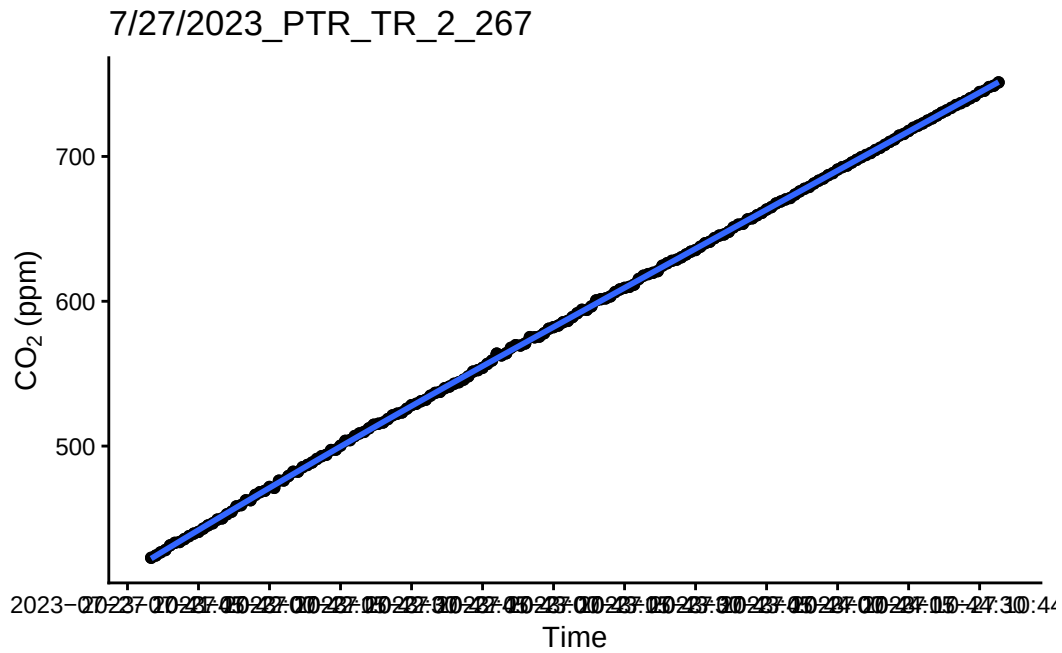
16 Look at the really high or really low ones to see if there is a timing issue or something

```
### pull out fluxes in the 5% and 95% tails to look at really high and really low
tail_plots <- unique(tails$Plot)

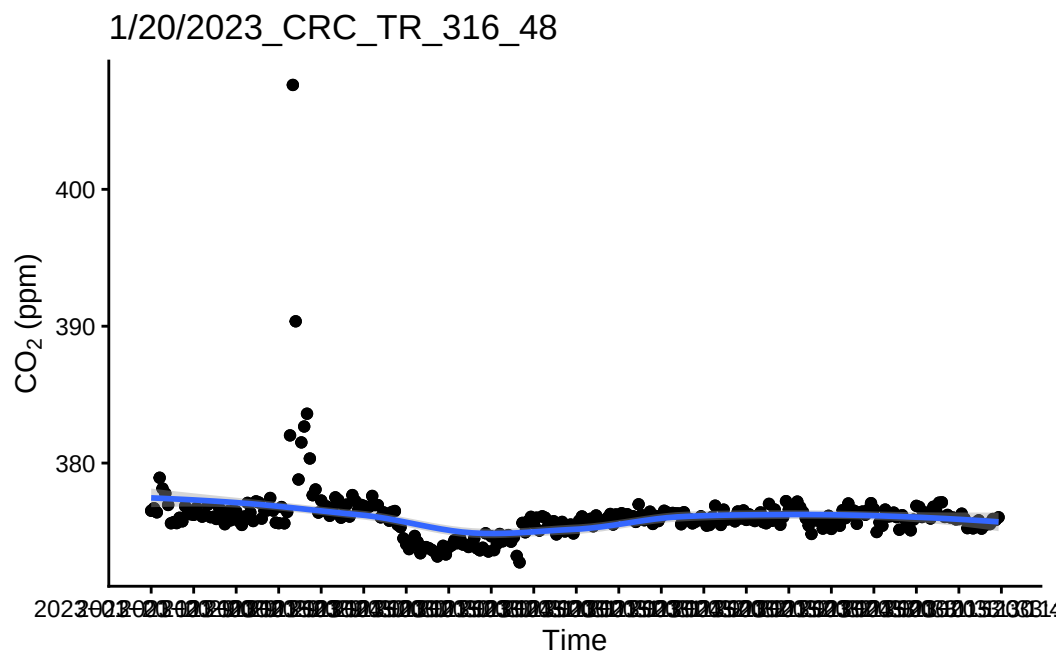
# take a random sample (FALSE so we don't get the same plot twice)
# set.seed(...) to guarantee reproducibility
tail_sample <- sample(tail_plots, 2, replace=FALSE)

#Loop to run five random plots subset above
for(P in tail_sample){
  r1 <- subset(alldat_trimmed, Plot == P & !is.na(metadata_row))
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +
    scale_x_datetime(breaks = "15 sec") +
    ylab(expression(paste( CO [2], " (ppm)"))) +
    xlab("Time") + labs(title = P) +
    geom_smooth()
  print(p1)
}
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

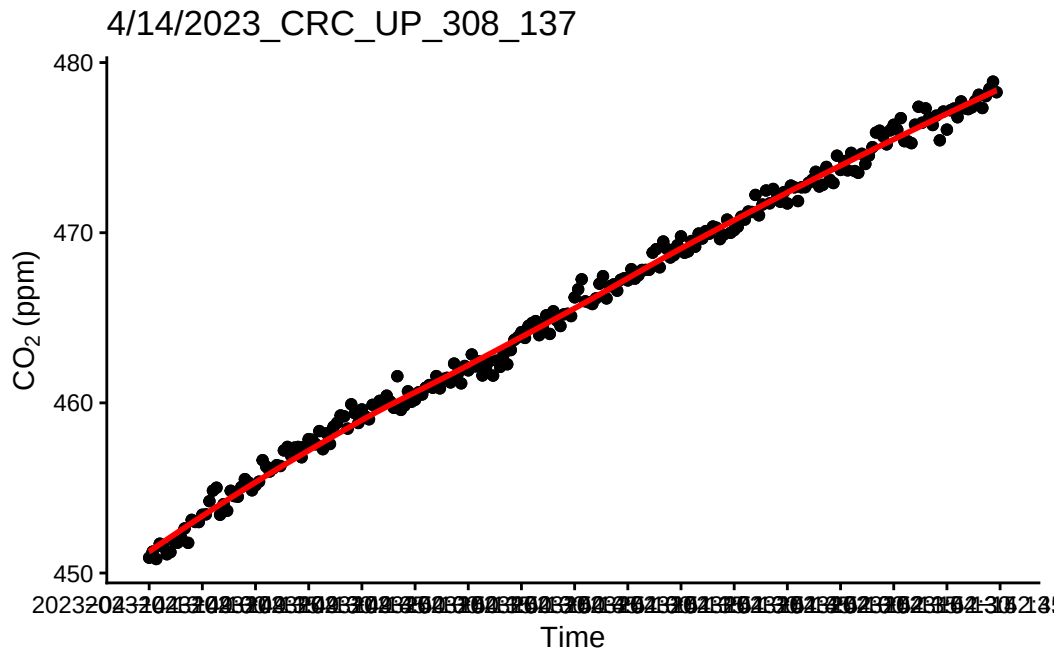


```
#would be nice to put on the plot the flux estimate and if it is a high or low or something  
#####
```

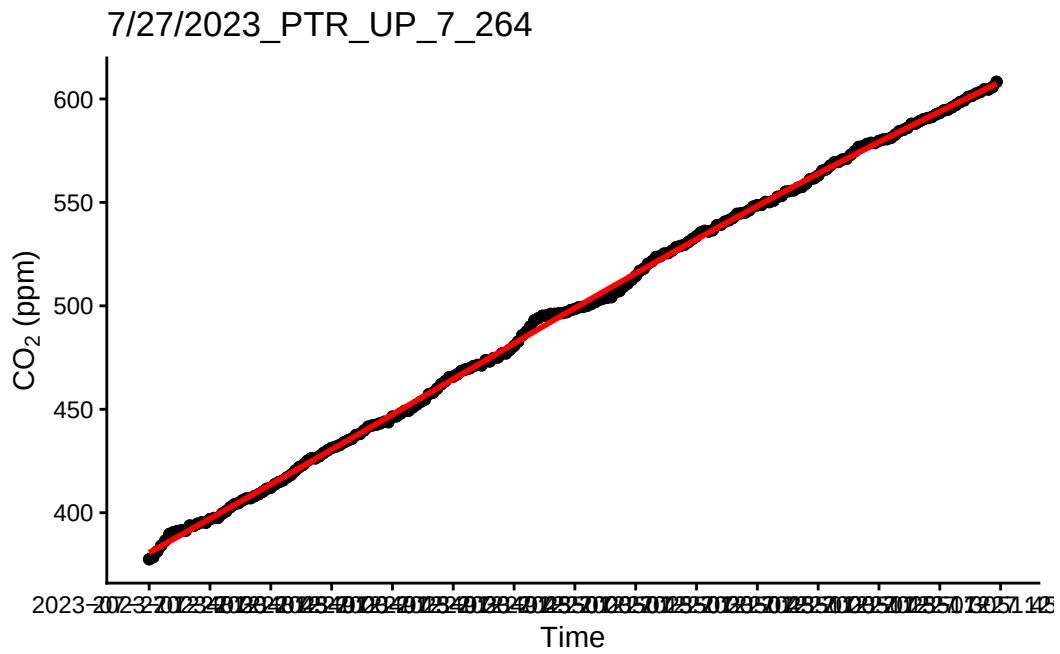
17 Check all fluxes

```
##Put all fluxes together  
#fluxes_all <- as.data.frame(subset(fluxes, select = Plot:TIMESTAMP_max))  
  
all_plots <- unique(fluxes$Plot)  
  
# take a random sample (FALSE so we don't get the same plot twice)  
# set.seed(...) to guarantee reproducibility  
all_sample <- sample(all_plots, 5, replace=FALSE)  
  
#Loop to run five random plots subset above  
for(P in all_sample){  
  r1 <- subset(alldat_trimmed, Plot== P & !is.na(metadata_row))  
  p1 <- ggplot(r1, aes(TIMESTAMP, CO2)) + geom_point() + theme_classic() +  
    scale_x_datetime(breaks = "15 sec") +  
    ylab(expression(paste( CO [2], " (ppm)"))) +  
    xlab("Time") + labs(title = P) +  
    geom_smooth(color="red")  
  print(p1)  
}
```

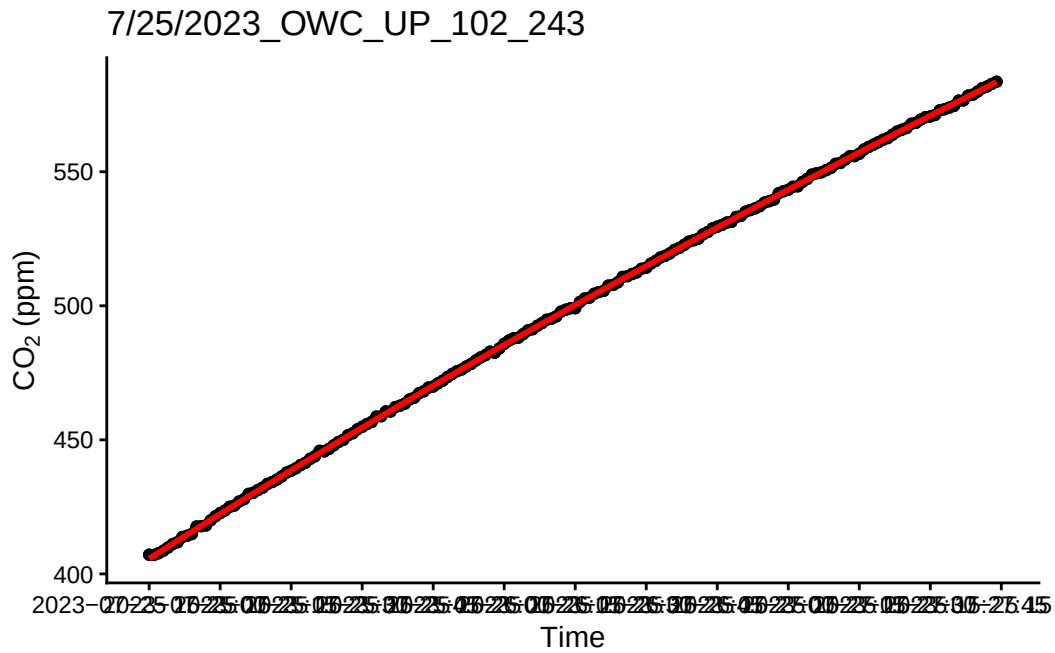
`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



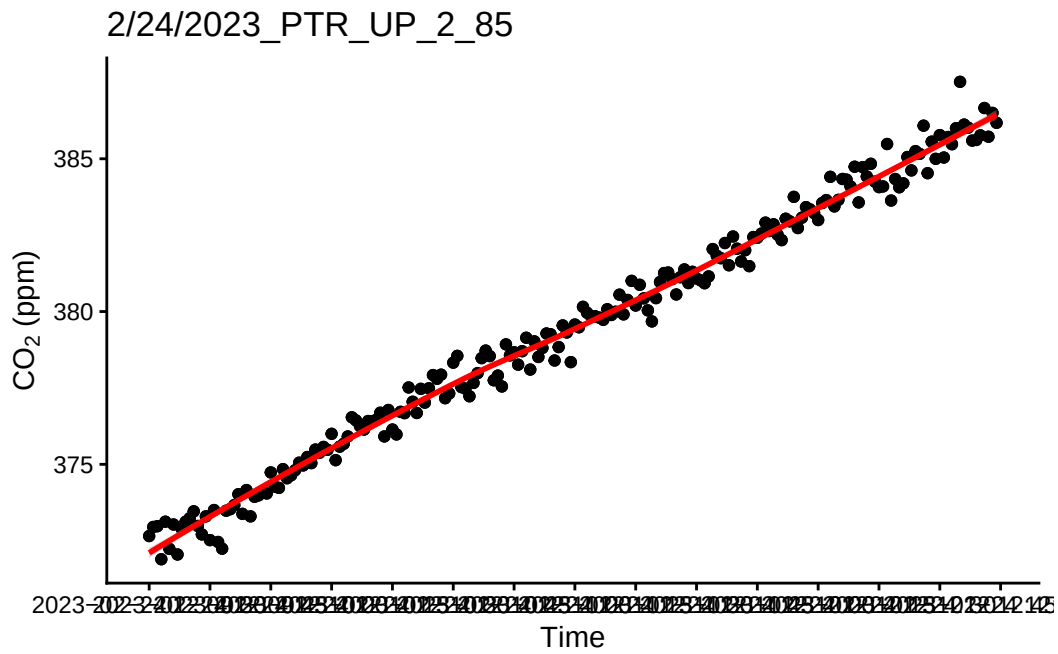
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



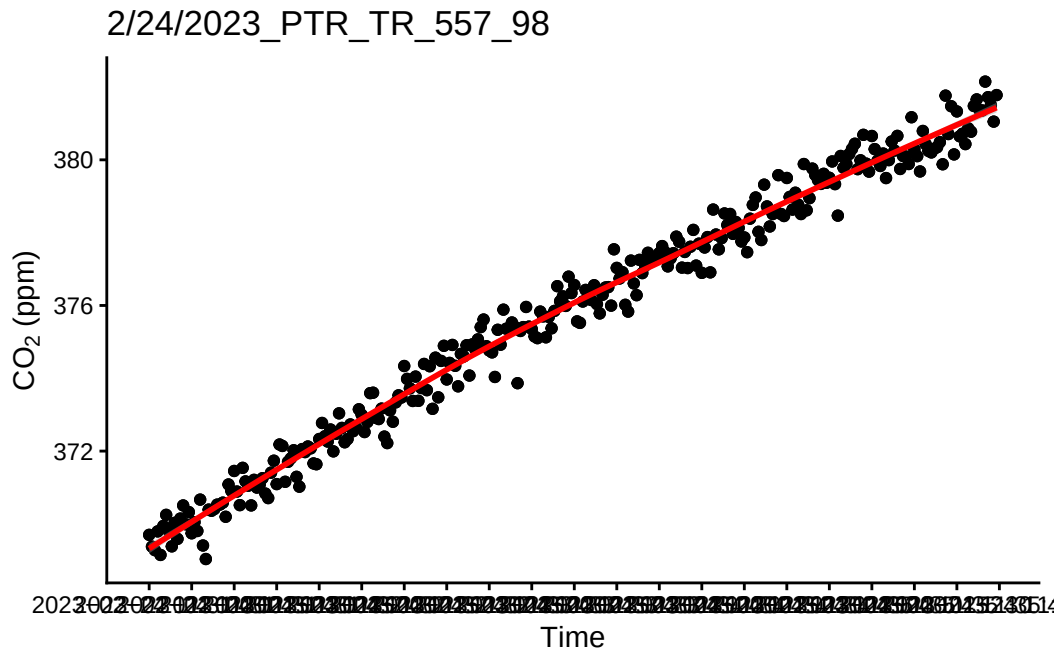
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



#####

18 QAQC - Flags

```
# QAQC flags
fluxes_flagged <- fluxes %>%
  mutate(
    # model fit flags
    CO2_lin_pass = (lin_r.squared >= 0.85 | lin_RMSE < 5) & !is.na(lin_r.squared),
    CO2_HM81_pass = (HM81_r.squared >= 0.85 | HM81_RMSE < 5) & !is.na(HM81_r.squared),
    CO2_poly_pass = (poly_r.squared >= 0.85 | poly_RMSE < 5) & !is.na(poly_r.squared),
    # negative flux flag
    CO2_neg_flux = lin_flux.estimate < 0 | is.na(lin_flux.estimate),
    # rejection flags
    CO2_rejected_model = !(CO2_lin_pass | CO2_HM81_pass | CO2_poly_pass),
    CO2_rejected_negflux = CO2_neg_flux,
    CO2_flux_rejected = CO2_rejected_model | CO2_rejected_negflux
  )

# percentage of problems
```

```

r2_summary <- fluxes_flagged %>%
  summarise(
    CO2_total_obs      = n(),
    CO2_n_rejected_model = sum(CO2_rejected_model, na.rm = TRUE),
    CO2_n_rejected_negflux = sum(CO2_rejected_negflux, na.rm = TRUE),
    CO2_n_rejected_both  = sum(CO2_rejected_model & CO2_rejected_negflux, na.rm = TRUE),
    CO2_n_rejected_total = sum(CO2_flux_rejected, na.rm = TRUE),
    CO2_pct_rejected_total = 100 * CO2_n_rejected_total / CO2_total_obs,
    CO2_n_lin_pass       = sum(CO2_lin_pass, na.rm = TRUE),
    CO2_n_HM81_pass      = sum(CO2_HM81_pass, na.rm = TRUE),
    CO2_n_poly_pass      = sum(CO2_poly_pass, na.rm = TRUE)
  )

# formatted table for PDF
r2_summary %>%
  tidyr::pivot_longer(everything(), names_to = "Metric", values_to = "Value") %>%
  mutate(
    Metric = recode(Metric,
      CO2_total_obs      = "Total observations",
      CO2_n_rejected_model = "Rejected - poor model fit (n)",
      CO2_n_rejected_negflux = "Rejected - negative CO2 flux (n)",
      CO2_n_rejected_both  = "Rejected - both reasons (n)",
      CO2_n_rejected_total = "Rejected - total (n)",
      CO2_pct_rejected_total = "Rejected - total (%)",
      CO2_n_lin_pass       = "Passed - Linear model (n)",
      CO2_n_HM81_pass      = "Passed - HM81 model (n)",
      CO2_n_poly_pass      = "Passed - Polynomial model (n)"
    ),
    Value = ifelse(Metric == "Rejected - total (%)",
      round(Value, 1),
      as.integer(Value)) ) %>%
  kable(
    caption = "QA/QC CO2 Flux Summary",
    align   = c("l", "r"),
    booktabs = TRUE,
    format   = "latex") %>%
  kable_styling(
    latex_options = c("striped", "hold_position"),
    full_width    = FALSE ) %>%
  pack_rows("Rejections", 2, 6) %>%
  pack_rows("Model performance", 7, 9)

```

Table 1: QA/QC CO2 Flux Summary

Metric	Value
Total observations	273.0
Rejections	
Rejected — poor model fit (n)	0.0
Rejected — negative CO2 flux (n)	5.0
Rejected — both reasons (n)	0.0
Rejected — total (n)	5.0
Rejected — total (%)	1.8
Model performance	
Passed — Linear model (n)	273.0
Passed — HM81 model (n)	202.0
Passed — Polynomial model (n)	273.0

```
# R2 vs RMSE diagnostic plot
# pivot_longer keeps R2/RMSE columns and uses CO2*_pass as the pass indicator
set.seed(42)
plot_data <- fluxes_flagged %>%
  slice_sample(n = min(15000, nrow(.))) %>%
  select(
    lin_r.squared, lin_RMSE, CO2_lin_pass,
    HM81_r.squared, HM81_RMSE, CO2_HM81_pass,
    poly_r.squared, poly_RMSE, CO2_poly_pass,
    CO2_rejected_negflux
  ) %>%
  # rename pass columns to match model prefix before pivoting
  rename(
    lin_pass = CO2_lin_pass,
    HM81_pass = CO2_HM81_pass,
    poly_pass = CO2_poly_pass
  ) %>%
  pivot_longer(
    cols = -CO2_rejected_negflux,
    names_to = c("model", ".value"),
    names_pattern = "(lin|HM81|poly)_(.*)"
  ) %>%
  rename(r2 = r.squared) %>%
  mutate(
    model = recode(model, lin = "Linear", HM81 = "HM81", poly = "Polynomial"),
```

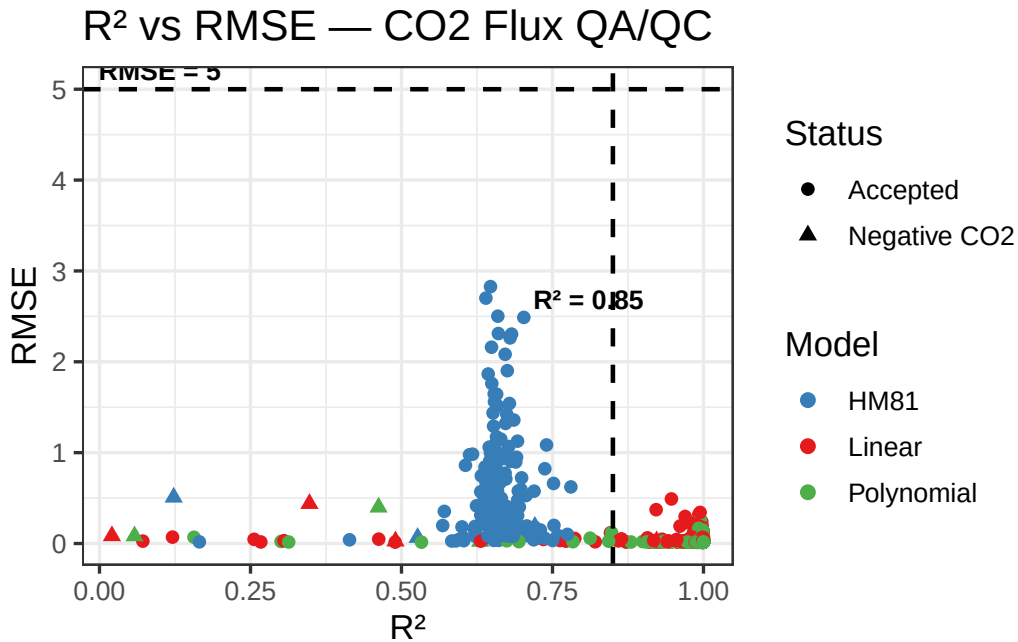
```

    status = case_when(
      CO2_rejected_negflux ~ "Negative CO2",
      pass                  ~ "Accepted",
      TRUE                 ~ "Rejected"
    )
  ) %>%
  filter(!is.na(r2), !is.na(RMSE))

#graphing
QAQC_plot <- ggplot(plot_data, aes(x = r2, y = RMSE, color = model, shape = status)) +
  geom_point(size = 2) +
  geom_vline(xintercept = 0.85, linetype = "dashed", color = "black", linewidth = 0.8) +
  geom_hline(yintercept = 5, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate("text", x = 0.7, y = max(plot_data$RMSE, na.rm = TRUE) * 0.95,
    label = "R2 = 0.85", hjust = -0.1, size = 3.5, fontface = "bold") +
  annotate("text", x = 0.1, y = 5,
    label = "RMSE = 5", vjust = -0.5, size = 3.5, fontface = "bold") +
  scale_color_manual(values = c(
    "Linear"      = "#E41A1C",
    "HM81"       = "#377EB8",
    "Polynomial" = "#4DAF4A"
  )) +
  scale_shape_manual(values = c(
    "Accepted"    = 16,
    "Rejected"    = 4,
    "Negative CO2" = 17 )) +
  labs(
    title = "R2 vs RMSE - CO2 Flux QA/QC",
    x      = "R2",
    y      = "RMSE",
    color  = "Model",
    shape  = "Status" ) +
  theme_bw(base_size = 13) +
  theme(legend.position = "right")

print(QAQC_plot)

```



19 Adding manual flag to reject any fluxes that do not fit criteria

```
# adding manual flags to the ones that are crazy off based on visual checks
manual_reject <- c("2/24/2023_PTR_UP_6_101" #leak
)

fluxes_flagged <- fluxes_flagged %>%
  mutate(
    CO2_rejected_manual = Plot %in% manual_reject,
    CO2_flux_rejected = CO2_rejected_model | CO2_rejected_negflux | CO2_rejected_manual)

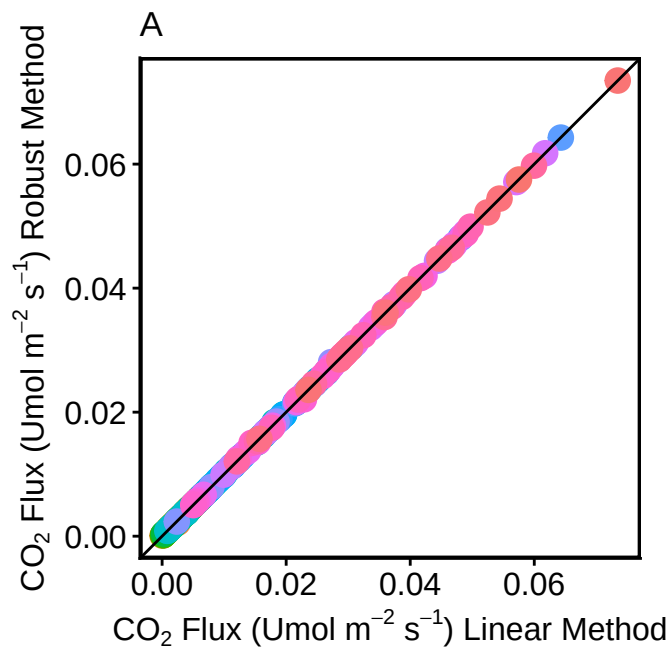
# fluxes_flagged %>%
#   filter(CO2_rejected_manual) %>%
#   select(Plot, lin_r.squared, lin_RMSE, CO2_rejected_model, CO2_rejected_negCO2, CO2_rejected_manual)
```

20 Do some visualization of QAQC plots to assess fluxes

```
allflux_filtered <- fluxes_flagged %>% filter(!CO2_flux_rejected)

p1 <- ggplot(allflux_filtered, aes(lin_flux.estimate, rob_flux.estimate, color = Plot)) +
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  labs(title = "A") +
  ylab(expression(paste(CO [2], " Flux (Umol m-2 s-1) Robust Method"))) +
  xlab(expression(paste(CO [2], " Flux (Umol m-2 s-1) Linear Method"))) +
  guides(color = guide_legend(title = "Plot")) +
  theme(legend.position="none") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
        aspect.ratio = 1)

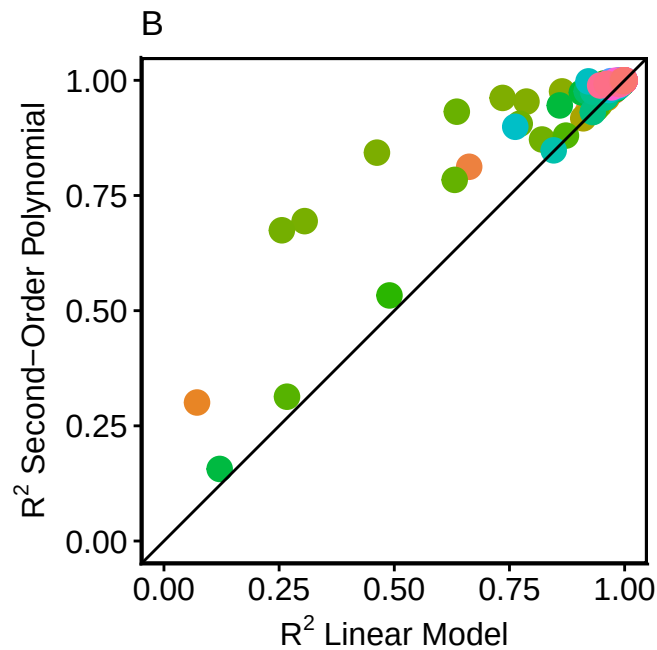
p1
```



```
p2 <- ggplot(allflux_filtered, aes(lin_r.squared, poly_r.squared, color = Plot)) +
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  xlim(0,1) + ylim(0,1) +
  labs(title = "B") +
```

```
ylab(expression(paste("R"2* " Second-Order Polynomial")))) +
xlab(expression(paste("R"2* " Linear Model")))) +
guides(color = guide_legend(title = "Plot_ID")) +
theme(legend.position="none") +
theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
      axis.title.y = element_text(size=12), legend.text=element_text(size=12),
      panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
      aspect.ratio = 1)
```

p2

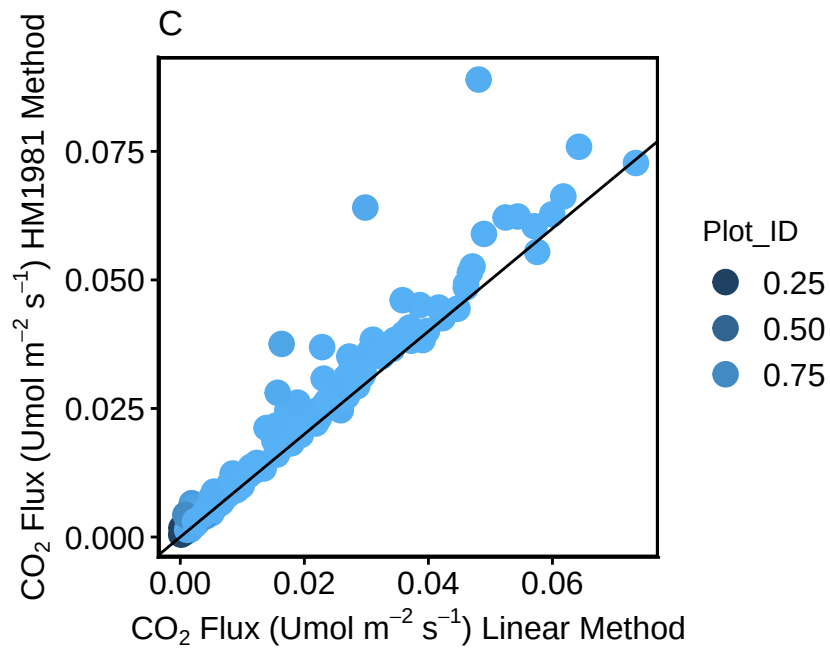


```
p3 <- ggplot(allflux_filtered, aes(lin_flux.estimate, HM81_flux.estimate, color = lin_r.squa
  geom_point(size=4) + geom_abline() +
  theme_classic() +
  #xlim(0,50) + #ylim(0,50) +
  labs(title = "C") +
  ylab(expression(paste( CO [2], " Flux (Umol m"-2* " s"-1*) HM1981 Method")))) +
  xlab(expression(paste( CO [2], " Flux (Umol m"-2* " s"-1*) Linear Method")))) +
  guides(color = guide_legend(title = "Plot_ID")) +
  theme(legend.position="right") +
  theme(axis.title.x = element_text(size=12), axis.text = element_text(size=12),
        axis.title.y = element_text(size=12), legend.text=element_text(size=12),
        panel.border = element_rect(colour = "black", fill=NA, linewidth =1),
```

```
aspect.ratio = 1)
```

p3

Warning: Removed 69 rows containing missing values or values outside the scale range (``geom_point()``).



```
#ggarrange(p1, p2, p3, ncol=3, nrow=1, common.legend = TRUE, legend="none")
```

21 Read out final dataframe and export for use elsewhere

```
allflux_filtered_renamed <- fluxes_flagged %>%
  mutate(
    Date = mdy(str_extract(Plot, "[0-9]+/[0-9]+/[0-9]{4}")),
    Month = month.abb[month(Date)],
    Site = str_extract(Plot, "(PTR|OWC|CRC)"),
    Zone = str_extract(Plot, "_TR_|_UP_") %>%
      str_remove_all("_"),
    Replicate = as.integer(
```

```

    str_extract(Plot, "(?<=_ (TR|UP)_)[0-9]+")
  ),
  Row_ID = as.integer(
    str_extract(Plot, "[0-9]+$")) ) %>%

# Optional: rebuild a clean Plot label
mutate(
  Plot_clean = paste(
    format(Date, "%Y-%m-%d"),
    Site,
    Zone,
    Replicate,
    Row_ID,
    sep = "_"
  )
) %>%

select(
  Plot_clean,
  Date,
  Month,
  Site,
  Zone,
  Replicate,
  Row_ID,
  everything(),
  -Plot
)

#export as a csv to the folder that you want
write.csv(allflux_filtered_renamed, file="Processed Data/COMPASS_Synoptic_LE_TreeStem_Fluxes.csv")
# #Change file name

```